

# EMILIANO DE LA GARZA VILLARREAL

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## PROFESSIONAL SUMMARY

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Data Science senior with specialized experience in Embedded Machine Learning and Robotics. Proven track record of architecting real-time CV pipelines on NVIDIA Jetson and optimizing trajectory prediction models using Federated Learning. Expert in ROS 2 and PyTorch, with a demonstrated ability to translate complex data into industrial cost-savings exceeding \$300,000 annually.

## EDUCATION

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**University of Tennessee - Chattanooga**  
*Bachelor of Science (BS), Data Science*

**August 2024 - May 2026**  
GPA: 3.8

**Cleveland State Community College**  
*Associate of Science (AS), Computer Science*

**August 2022 - December 2023**  
GPA: 3.7

## PROFESSIONAL EXPERIENCE

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**Center of Urban Informatics and Progress**  
*Assistant Researcher*

**April 2025 - Present**  
Chattanooga, TN, USA

- Built a real-time intersection monitoring system on NVIDIA Jetson (Embedded System) using ROS 2; optimized performance by architecting an asynchronous multi-worker pipeline to maximize GPU utilization and eliminate thread-blocking latency for trajectory visualization.
- Improved Average Displacement Error (ADE) and Final Displacement Error (FDE) metrics by ~8% over baseline strategies by applying and combining FedProx and FedAdam within a Flow-based trajectory prediction model.
- Managed the applied machine learning lifecycle for a crash detection pipeline using Multiscale Vision Transformers (MViTv2), encompassing video dataset curation, spatio-temporal transformations, and local model training.

**Eaton Corporation**  
*Engineering Intern*

**December 2023 - April 2026**  
Cleveland, TN, USA

- Analyzed large-scale industrial datasets to identify operational trends and inefficiencies, translating raw metrics into actionable insights to generate over \$300,000 in annual savings.
- Conducted shop-floor time studies to build a predictive production model, updating system standards that reduced labor hours by 20%.

## PROJECTS & OUTSIDE EXPERIENCE

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### Swarm-Sense (ROS 2 & Distributed Systems)

- Architected a high-performance ROS 2 Humble fleet interface to monitor 5+ robots simultaneously via Foxglove Bridge and Server-Client WebSockets.
- Reduced video latency to sub-100ms by implementing hardware-accelerated H.264 transport and 500kbps bitrate caps for bandwidth-constrained environments.
- Developed a custom TypeScript/React extension for edge-side telemetry caching and automated data export to MATLAB (.mat) formats.

## SKILLS

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**Machine Learning & Computer Vision:** PyTorch, YOLO, OpenCV, Federated Learning, Transfer Learning, Time-Series Analysis, Data Augmentation, Neural Networks, Machine Learning, Computer Vision

**Edge AI & Robotics Systems:** NVIDIA Jetson, TensorRT, ROS2, Real-Time Data Pipelines, RTSP

**Programming & Core Tools:** Python, C/C++, Linux/Unix, Git

**Data Analytics & Operations:** Pandas, NumPy, Data Engineering, Data Analysis, Operational Analysis, Shop-Floor Time Studies

**Languages:** Spanish, English